Kamarina Hewett
Homework # 4

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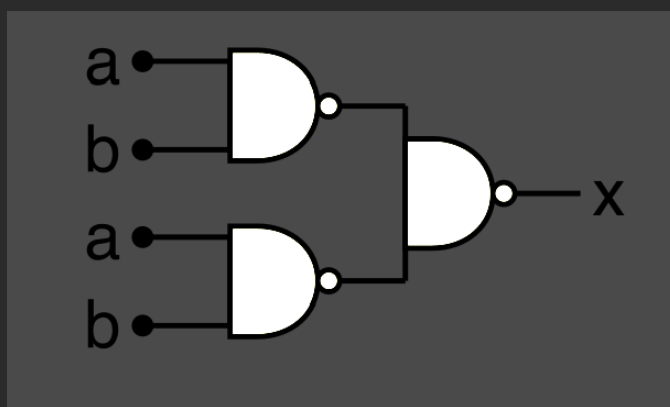
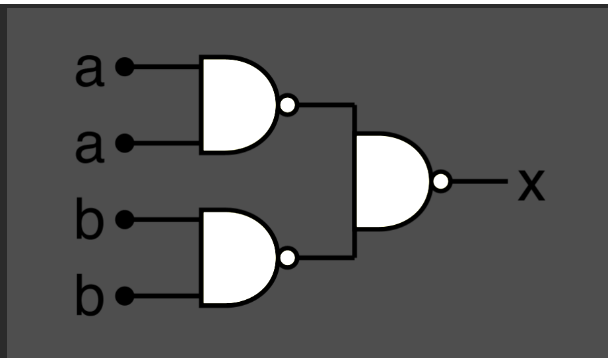
#Please write four Python3 scripts that create the following using waveDROM:
# #1. An AND gate made from NANDs
import wavedrom
svg = wavedrom.render("""
{ "assign":[
 ["x", ["~&",
   ["~&", "a", "b" ],
   ["~&", "a", "b" ]
 ]]
]}""")
svg.saveas("HW4_ANDfromNAND.svg")
```

Picture:




```
############################################################
#2. An OR gate made from NANDs
import wavedrom
svg = wavedrom.render("""
{ "assign":[
 ["x", ["~&",
   ["~&", "a" , "a"],
   ["~&", "b" , "b"]
 ]]
]}""")
svg.saveas("HW4_ORfromNAND.svg")
```

Picture (on next page):




```
###############################################################
#3. an XOR gate made from NANDs
import wavedrom
svg = wavedrom.render("""
{ "assign":[
 ["x",
   ["~&",
       ["~&",
           ["=","a"],
           ["~&" , "a" , "b" ]
       ],
       ["~&",
           ["~&" , "a" , "b" ],
           ["=","b"]
       ]
   ]
 ]
]}""")
svg.saveas("HW4_XORfromNAND.svg")
```

Picture: